「Def.

Given four complex numbers $a, b, c, d \in \mathbb{C}$ with $ad-bc \neq 0$, a map

$$w = T(z) = \frac{a \cdot z + b}{c \cdot z + d}$$

is called a linear fractional transformation.

「observation for the linear fraction.

The condition $ad-bc\neq 0$ gives that the $T(z)$ is non-constant, assume $ad-bc=0$.

If $c \neq 0$, then

$$\frac{az+b}{cz+d} = \frac{\frac{a}{c}\cdot(cz+d)+b-\frac{ad}{c}}{cz+d} = \frac{a}{c}.$$

If $a \neq 0$, then

$$\frac{az+b}{cz+d} = \frac{az+b}{cz+\frac{bc}{a}} = \frac{a}{c}\cdot\left(\frac{z+\frac{b}{a}}{z+\frac{b}{a}}\right) = \frac{a}{c} \quad \left(z \neq -\frac{b}{a}\right).$$

If $a=0$, then either $b=0$ or $c=0$.

If $b=0$, then $T\equiv 0$.

If $c=0$, then $T \equiv \frac{b}{d}$

If $c=0$, then either $a=0$ or $d=0$.

If $a=0$, then $T\equiv\frac{b}{d}$

If $d=0$, then T is not defined.

Now, we assume $ad-bc\neq 0$, then on $\mathbb{C}\setminus\{-\frac{d}{c}\}$.

$$w = T(z) = \begin{cases} \frac{a}{c} - \left[\frac{ad-bc}{c^2}\right]\cdot\frac{1}{z+\frac{d}{c}} & \text{if } c\neq 0 \\ \frac{a}{d}z + \frac{b}{d} & \text{if } c=0, \end{cases}$$

Therefore, the T is injective.

「Thm.

Define $\mathbb{H} \overset{\text{def}}{=} \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}$. (That is, the upper plane)

A map $F : \mathbb{H} \to \mathbb{D} : z \mapsto \dfrac{i-z}{i+z}$ is Conformal map,

and $\quad G : \mathbb{D} \to \mathbb{H} : w \mapsto i \cdot \dfrac{1-w}{1+w}$ is the inverse of F.

Proof. Given $a + i b \in \mathbb{H}$ ($b > 0$),

$$\|(0,1) - (a,b)\|^2 = a^2 + (b-1)^2 < \|(0,-1) - (a,b)\|^2 = a^2 + (b+1)^2.$$

Therefore for all $z \in \mathbb{H}$, $|F(z)| < 1$, so the F is well-defined

And, given $u + iv \in \mathbb{D}$ ($u^2 + v^2 < 1$),

$$\operatorname{Im}(G(w)) = \operatorname{Re}\left[\frac{1-(u+iv)}{1+u+iv}\right] = \operatorname{Re}\left[\frac{(1-u-iv)(1+u-iv)}{(1+u)^2 + v^2}\right]$$

$$= \frac{1-u^2-v^2}{(1+u)^2+v^2} > 0$$

Therefore the G is also well-defined. And, these are rational of polynomials, So holomorphic. Now, given $z \in \mathbb{H}$,

$$G(F(z)) = i \cdot \frac{1 - \dfrac{i-z}{i+z}}{1 + \dfrac{i-z}{i+z}} = i \cdot \frac{i+z-i+z}{i+z+i-z} = z,$$ and similarly $F \circ G = id$.

So, these are bijection with $G = F^{-1}$

Observe that for a real number $x \in \mathbb{R}$,

$$F(x) = \frac{i-x}{i+x} = \frac{(i-x)(-i+x)}{1+x^2} = \frac{1+2xi-x^2}{1+x^2} = \frac{1-x^2}{1+x^2} + i \cdot \frac{2x}{1+x^2},$$

and Parametrize $x = \tan t$ ($-\frac{\pi}{2} < t < \frac{\pi}{2}$), then

$$\sin 2a = 2 \cdot \sin a \cdot \cos = \frac{2\tan a}{1+\tan^2 a} \quad \text{and} \quad \cos 2a = \frac{1-\tan^2 a}{1+\tan^2 a}, \quad \text{so}$$

$$F(x) = \cos 2t + i \cdot \sin 2t = e^{2it}.$$